Exact reproduction and KS-atlas transfer of a Morales–Ramis obstruction near the parabolic Lagrange solution

Jacob Sullivan
Affiliation and ORCID to be confirmed before submission

24 July 2026

Abstract

We give an exact, machine-qualified reproduction of the planar Morales–Ramis obstruction near the parabolic Lagrange phase curve for every finite positive mass triple. Starting from the declared translation- and rotation-reduced Hamiltonian, an implementation disjoint from the production certificate differentiates the Hamiltonian, verifies the phase curve, constructs the full six-dimensional variational equation, performs the Whittaker energy-shell reduction, and derives a four-dimensional Fuchsian normal variational equation. Its finite residue spectra and infinity characteristic polynomial imply that the differential-Galois identity component is non-Abelian throughout the positive-mass domain. We then transfer that local obstruction to the translation-reduced spatial Hamiltonian by an explicit variational subquotient and to a qualified two-chart Kustaanheimo–Stiefel (KS) atlas by exact cotangent reduction, fixed-energy Sundman equivalence, and a shell-relative rank argument. The resulting theorem concerns complete meromorphic Liouville integrability locally near one fixed-nonzero-angular-momentum phase curve. It does not exclude one additional integral or make a global assertion. The planar theorem is a reproduction of published work, not an asserted original discovery. All decision-bearing formulas, source hashes, negative controls, and qualification records are included in a one-command reproducibility package.

1 Scope and prior art

Differential Galois theory supplies necessary conditions for the meromorphic Liouville integrability of Hamiltonian systems. In the form used here, the Morales–Ramis theorem requires the identity component of the differential Galois group of the variational equation, and hence of the relevant normal variational equation, to be Abelian near a non-equilibrium phase curve [5]. Tsygvintsev applied this framework to the parabolic Lagrange solution of the planar three-body problem and derived the Fuchsian normal system reproduced below [6]. Boucher and Weil gave an independent published application at the theorem level [1].

The scientific contribution qualified in this paper has four deliberately separated parts:

1. an exact project-owned reconstruction of the planar normal variational equation from the reduced Hamiltonian;

2. a source-equivalence audit against hash-pinned primary papers;

3. an exact transfer of the obstruction to the spatial variational module; and

4. an exact transfer to a previously qualified multi-chart KS regularization.

The planar nonintegrability result itself is prior art. We do not claim its discovery, its first proof, or novelty of its mathematical conclusion.

The word "integrability" will always mean complete Liouville integrability by meromorphic first integrals on a complex neighborhood unless another meaning is explicitly stated. For the translation-reduced spatial Hamiltonian, which has six degrees of freedom, completeness requires six functionally independent commuting integrals including the Hamiltonian. The result is local to the neighborhood of a specified phase curve. It is not an assertion about every trajectory or every phase-space domain.

1.1 Models and normalizations

All masses are finite and strictly positive. In the planar reproduction we set the gravitational constant and the fixed angular-momentum scale to one, and normalize the third mass to $m_3 = 1$. Positive common scaling recovers every positive mass triple. The spatial theorem is stated on the regular collision-free portion of the translation-reduced phase space. Binary collision extensions enter only through restriction back to that open set; triple collision is outside the qualified model.

2 Reduced planar model

Let

$$M_1 = 1 + \frac{1}{m_1}, \qquad M_2 = 1 + \frac{1}{m_2}, \qquad P = p_3 q_2 - p_2 q_3 - 1.$$

The declared reduced Hamiltonian is

$$\begin{aligned} H = {} & \frac{M_1}{2}\left(p_1^2 + \frac{P^2}{q_1^2}\right) + \frac{M_2}{2}(p_2^2 + p_3^2) + p_1 p_2 - \frac{p_3 P}{q_1} \\ & - \frac{m_1}{q_1} - \frac{m_2}{\sqrt{q_2^2 + q_3^2}} - \frac{m_1 m_2}{\sqrt{(q_1 - q_2)^2 + q_3^2}}. \end{aligned} \tag{1}$$

The implementation reconstructs every derivative used below from (1); it does not import a published variational matrix.

Put

$$S_1 = m_1 + m_2 + 1, \qquad S_2 = m_1 m_2 + m_1 + m_2, \qquad S_3 = m_2 + 2.$$

The equilateral parabolic phase curve is parametrized by

$$\begin{aligned} q_1 &= q, & q_2 &= \frac{q}{2}, & q_3 &= \frac{\sqrt{3}\, q}{2}, \\ p_1 &= p, & p_2 &= Ap + \frac{B}{q}, & p_3 &= Cp + \frac{D}{q}, \end{aligned} \tag{2}$$

where

$$\begin{aligned} A &= \frac{m_2(1 - m_1)}{m_1 S_3}, & B &= -\frac{\sqrt{3}\, S_1 m_2}{S_2 S_3}, \\ C &= \frac{\sqrt{3}\, m_2(m_1 + 1)}{m_1 S_3}, & D &= -\frac{m_2(S_2 + m_1 m_2 - 1)}{S_2 S_3}. \end{aligned}$$

On the zero-energy surface let $w = pq$. Then

$$q(w) = -\frac{aw^2 + bw + d}{c}, \qquad p(w) = \frac{w}{q(w)}, \tag{3}$$

with

$$a = \frac{2S_1S_2}{m_1^2S_3^2}, \quad b = -\frac{2\sqrt{3}\,m_2S_1}{m_1S_3^2}, \quad c = -S_2, \quad d = \frac{2S_1(m_2^2 + m_2 + 1)}{S_3^2S_2}.$$

Proposition 1 (Exact phase-curve check)**.** *For every positive m_1, m_2, equations (2)–(3) define a non-equilibrium zero-energy integral phase curve of (1) away from its isolated singular parameter values.*

Proof. Direct substitution gives $H = 0$. The exact reference calculation also verifies that all six components of $J_6\nabla H$ equal the corresponding components of dz/dw multiplied by one common rational expression dw/dt. This stronger tangent identity rules out the weaker possibility that the displayed parametrization merely lies on the energy shell. □

3 Exact normal variational equation

The full planar variational equation along the phase curve is

$$\frac{d\,\delta z}{dt} = J_6H_{zz}(\Gamma(w))\,\delta z.$$

The reference implementation forms this 6 by 6 matrix by exact differentiation and verifies its Hamiltonian-matrix identity.

To remove the energy/flow pair, solve $H = 0$ locally for p_1 and put $K = -p_1$. For $y = (q_2, q_3, p_2, p_3)$, implicit differentiation gives

$$K_{ij} = \frac{H_{ij}}{H_p} - \frac{H_{ip}H_j}{H_p^2} - \frac{H_iH_{pj}}{H_p^2} + \frac{H_iH_{pp}H_j}{H_p^3}, \tag{4}$$

where $p = p_1$. After substituting the phase curve and changing the independent variable from q_1 to w, the four-dimensional normal system is

$$A_{\mathrm{NVE}} = J_4\,\frac{dq}{dw}\,\mathrm{Hess}_y(K). \tag{5}$$

Symmetry of the Hessian and the Hamiltonian identity for (5) are exact qualification gates.

Define

$$\begin{aligned} L &= 2S_2w - \sqrt{3}\,m_1m_2, \\ Z &= S_2^2w^2 - \sqrt{3}\,m_1m_2S_2w + m_1^2(m_2^2 + m_2 + 1), \\ C &= \mathrm{diag}(LZ, LZ, 1, 1). \end{aligned} \tag{6}$$

The gauge-transformed system is

$$A_F = C^{-1}A_{\mathrm{NVE}}C - C^{-1}\frac{dC}{dw}.$$

Exact denominator valuations show that it is Fuchsian, with three distinct finite singularities

$$t_0 = \frac{\sqrt{3}\, m_1 m_2}{2S_2},$$
$$t_1 = \frac{m_1(\sqrt{3}\, m_2 + iS_3)}{2S_2}, \qquad t_2 = \overline{t_1}. \tag{7}$$

The residues are computed from the generated matrix. Their characteristic polynomials are

$$\chi_{t_0}(\lambda) = \lambda^2(\lambda+1)^2,$$
$$\chi_{t_1}(\lambda) = \chi_{t_2}(\lambda) = \lambda(\lambda-1)(\lambda+1)(\lambda+2). \tag{8}$$

The residue at infinity is minus the finite-residue sum, and

$$\chi_\infty(\lambda) = \lambda^4 - 6\lambda^3 + 7\lambda^2 + 6\lambda + (27\sigma - 8), \qquad \sigma = \frac{S_2}{S_1^2}. \tag{9}$$

These identities are obtained in one generic exact computation over $\mathbb{Q}(\sqrt{3}, i)(m_1, m_2)$. Specialized rational mass cases are regressions, not substitutes for the generic derivation.

4 Positive-mass Galois obstruction

Before the normalization $m_3 = 1$,

$$\sigma = \frac{m_1m_2 + m_2m_3 + m_3m_1}{(m_1 + m_2 + m_3)^2}.$$

Positivity gives $\sigma > 0$, while

$$\frac{1}{3} - \sigma = \frac{(m_1 - m_2)^2 + (m_2 - m_3)^2 + (m_3 - m_1)^2}{6(m_1 + m_2 + m_3)^2} \geq 0. \tag{10}$$

Thus $0 < \sigma \leq 1/3$, with equality only at equal masses.

Lemma 2 (Non-Abelian identity component)**.** *For every finite positive mass triple, the identity component of the differential Galois group of the normal variational equation* (5) *is non-Abelian.*

Proof. The spectra in (8) are integral, so every finite local monodromy has all eigenvalues equal to one and is unipotent. Each unipotent element of a complex linear algebraic group lies in its identity component. Assume that identity component is Abelian. The finite monodromies then commute, and their product is unipotent. The fundamental-group relation would make the monodromy at infinity unipotent as well.

The infinity exponents obtained from (9) obey

$$\big((2\lambda - 3)^2 - 13\big)^2 = 144(1 - 3\sigma). \tag{11}$$

Unipotent infinity monodromy would require all four exponents to be integers. For two nested-radical roots put $x = 2\lambda_1 - 3$ and $y = 2\lambda_2 - 3$. Then x, y are odd integers with $x^2 + y^2 = 26$. The only odd-square decomposition is $1 + 25$, which forces the inner square root in (11) to be 12 and hence $\sigma = 0$. This contradicts (10). Therefore infinity monodromy is not unipotent and the assumed Abelian identity component is impossible. □

Proposition 3 (Reproduced planar obstruction)**.** *For every finite positive mass triple, the declared translation- and rotation-reduced planar Newtonian Hamiltonian at the fixed nonzero angular momentum is not completely integrable by meromorphic first integrals in a complex neighborhood of the parabolic Lagrange phase curve.*

Proof. Complete meromorphic Liouville integrability would, by the Morales–Ramis necessary condition, make the identity component of the normal variational differential Galois group Abelian [5]. Lemma 2 gives the contradiction. □

The first-normal-variational polynomial-invariant values $\{1/3, 8/27, 2/9\}$ are not exceptions to this complete-integrability obstruction. The positive-mass exceptional set for Proposition 3 is empty.

5 Spatial and KS-atlas transfer

5.1 Spatial variational subquotient

Let U be the collision-free regular part of the twelve-dimensional translation-reduced spatial phase space. In pair coordinates its Hamiltonian has the form

$$H_{\mathrm{sp}} = \frac{|W_1|^2}{2\mu_1} + \frac{|W_2|^2}{2\mu_2} + \frac{W_1 \cdot W_2}{m_c} - \frac{k_1}{|X_1|} - \frac{k_2}{|X_2|} - \frac{k_3}{|X_1 - X_2|}. \tag{12}$$

Reflection across the physical plane is a symplectic involution preserving (12). Along a planar solution, the in-plane and vertical variational subspaces are invariant. The in-plane block is the translation-reduced planar variational equation.

The planar rotation momentum is

$$L_z = X_{1x}W_{1y} - X_{1y}W_{1x} + X_{2x}W_{2y} - X_{2y}W_{2x}.$$

Rotational invariance gives $\{H_{\mathrm{sp}}, L_z\} = 0$. At the regular fixed nonzero momentum of the phase curve, the invariant tangent space $\ker dL_z$, quotiented by the rotation generator X_{L_z}, gives the reduced planar variational equation [4]. Energy restriction and flow quotient then give the four-dimensional normal module:

$$\begin{aligned}\text{spatial VE } 12 &\longrightarrow \text{planar submodule } 8 \longrightarrow \text{fixed-}L_z \text{ tangent } 7\\ &\longrightarrow \text{rotation quotient } 6 \longrightarrow \text{energy tangent } 5 \longrightarrow \text{normal quotient } 4.\end{aligned}$$

The Galois representation of a differential subquotient is an algebraic image of the full representation. An Abelian identity component has only Abelian algebraic images. Lemma 2 therefore makes the identity component of the spatial variational Galois group non-Abelian.

5.2 KS cotangent reduction

For one KS spinor $q = (q_1, q_2, q_3, q_4)$, use the convention [2]

$$\begin{aligned}\kappa(q) &= \big(2(q_1q_3 + q_2q_4), 2(q_1q_4 - q_2q_3), q_1^2 + q_2^2 - q_3^2 - q_4^2\big),\\ r &= q \cdot q, \qquad Y(q) = (-q_2, q_1, -q_4, q_3).\end{aligned} \tag{13}$$

The exact identities

$$|\kappa(q)|^2 = r^2, \quad D\kappa D\kappa^T = 4r\mathrm{I}_3, \quad D\kappa^T D\kappa = 4r\mathrm{I}_4 - 4YY^T, \quad D\kappa\, Y = 0$$

show that the configuration kernel is the circle direction generated by Y. The cotangent momentum map and physical projection are

$$\Xi(q,P) = Y(q)\cdot P, \qquad x = \kappa(q), \qquad p = \frac{D\kappa(q)P}{4r}. \tag{14}$$

Direct differentiation gives canonical projected brackets on $\Xi = 0$; away from that level, the momentum brackets have the expected magnetic terms proportional to Ξ. Applying (14) to two spinors gives the regular dimension chain

$$18 \longrightarrow 16 \longrightarrow 14$$

for zero-level restriction and quotient by $\mathbb{T}^2$, including the temporal canonical pair.

The regularized extended Hamiltonian is

$$\mathcal{K} = g(H_{\mathrm{sp}} + p_t), \qquad g = r_{12}r_{23}r_{31}, \qquad \frac{dt}{ds} = g. \tag{15}$$

For a projectable function F with no explicit t-dependence,

$$\{F,\mathcal{K}\} = g\{F,H_{\mathrm{sp}}\} + (H_{\mathrm{sp}} + p_t)\{F,g\}.$$

On $\mathcal{K} = 0$, with $g \neq 0$, conservation for $\mathcal{K}$ is equivalent to conservation for H_{sp}. No off-shell equivalence is claimed.

Definition 4 (Physically admissible KS family)**.** A complete candidate family consists of

$$(\mathcal{K}, p_t, \Xi_1, \Xi_2, F_1, \ldots, F_5).$$

It must be pairwise commuting, have rank nine on a relative dense-open subset of $\{\mathcal{K} = \Xi_1 = \Xi_2 = 0\}$, be invariant under both circle actions, agree on qualified chart overlaps, and be independent of the regularizing clock.

The shell-relative rank condition is essential. In adapted canonical coordinates, a candidate containing $\Xi_1 p_6$ can have ambient rank nine away from $\Xi_1 = 0$ while losing its physical direction on the shell. The qualification contains this exact negative control.

Lemma 5 (KS shell transfer)**.** *Every physically admissible KS family descends on the regular shell to six independent commuting physical functions* $(H_{\mathrm{sp}}, f_1, \ldots, f_5)$.

Proof. Commutation with p_t, Ξ_1, Ξ_2 removes the temporal-angle components of dF_i. On the shell write

$$dF_i = A_i + b_i\, dp_t + c_i^a\, d\Xi_a, \qquad d\mathcal{K} = g(dH_{\mathrm{sp}} + dp_t) + k^a\, d\Xi_a.$$

If the descended covectors $dH_{\mathrm{sp}}, d(A_i - b_i H_{\mathrm{sp}})$ were dependent, the displayed identities would lift that dependence to one among

$$d\mathcal{K},\ dp_t,\ d\Xi_1,\ d\Xi_2,\ dF_1, \ldots, dF_5,$$

contradicting shell-relative rank nine. The Poisson product rule from (15) similarly transfers conservation and involution. Circle invariance and overlap equality make the descended functions single-valued on the physical quotient. □

Theorem 6 (Qualified spatial and KS obstruction)**.** *For every finite positive mass triple, the complexified translation-reduced spatial Newtonian three-body Hamiltonian is not completely integrable by meromorphic first integrals in a neighborhood of the embedded parabolic Lagrange phase curve at fixed nonzero angular momentum. Consequently, the qualified multi-chart KS atlas admits no complete shell-regular physically admissible meromorphic Liouville family on any domain containing the regular lift of that neighborhood.*

Proof. The normal module in Lemma 2 is a differential subquotient of the spatial variational module. The Morales–Ramis condition therefore rules out complete local meromorphic Liouville integrability of the spatial Hamiltonian. If a physically admissible KS family existed on a domain containing the regular lifted neighborhood, Lemma 5 would descend its restriction to a complete physical family there, contradicting the spatial obstruction. □

The KS consequence requires shell regularity, zero Ξ, gauge projectability, chart compatibility, and clock independence. Omitting any one of these conditions changes the model and is outside Theorem 6.

6 Computational qualification

The proof package separates scientific inputs, production certificates, and a source-disjoint symbolic reconstruction. Each predecessor is accepted only by qualification identifier and exact SHA-256 hash. Status-only records are rejected. The principal records are summarized in Table 1.

Campaign	Role	Qualified status
WP2-MULTIQ-v1	Two-chart KS model and transitions	PASS
WP4-MR-REPL-v1	Generic planar NVE reproduction	PASS
WP4-LIFT-NI-v3	Spatial and KS transfer	PASS
WP1-CAT-N3BP-v1	External periodic-orbit catalogue validation	PASS
WP1-RESIDUAL-MAP-v1	Follow-on residual allocation	PASS
WP3-MULTI-CAND-v2-KQ-v1	Frozen-library bounded null search	BOUNDED NULL

Table 1: Qualified inputs. The last three rows provide computational context only and are not premises of Theorem 6.

The generic symbolic replay performs the following decision-bearing checks:

1. exact zero-energy and Hamilton-equation tangency of the phase curve;
2. construction and Hamiltonian identities of the six- and four-dimensional variational systems;
3. symmetry of the Whittaker Hessian;
4. Fuchsian pole support and simple-pole valuations;
5. exact finite and infinity characteristic polynomials;
6. the positive-mass sum-of-squares identity; and
7. byte-independent equality of the complete normalized result payload.

Elapsed time is excluded from the payload comparison. Four exact rational mass cases and a dependency-free production implementation provide additional convention and regression checks.

Adversarial controls mutate residue polynomials, source hashes, theorem wording, phase-curve scope, and prerequisite records. They also attempt to promote bounded discovery evidence into a theorem premise, accept an ambient-only KS rank family, accept an internal or anonymous review as independent, and authorize submission with incomplete metadata. Every such mutation must be rejected.

The broader legacy suite is partitioned into deterministic fast tests, slow simulations, and explicitly quarantined historical prototypes. That suite is software-regression evidence only; it does not establish the theorem.

7 Bounded discovery context

The WP1 catalogue track validates the project orbit pipeline against an external unequal-mass periodic-orbit dataset derived from the catalogue of Li, Li, and Liao [3]. This supports the numerical efficacy of the orbit tooling under a frozen convention and tolerance contract. It is not evidence for the differential-Galois conclusion.

WP3 searched a frozen library of 285 candidate functions after quotienting the known L^2 class. No promoted invariant survived the locked qualification. The WP3 result is bounded to its frozen library and is not used in the proof. It does not constitute a search over all functions and does not support an absence-of-integrals claim.

The two tracks answer different questions. WP1 tests whether the numerical pipeline reproduces external orbit evidence. WP3 allocates a finite candidate search. Theorem 6 instead follows from an exact local normal-variational obstruction.

8 Limitations and nonclaims

The theorem concerns complete meromorphic Liouville integrability locally near one phase curve. The result does not exclude one additional integral. It does not exclude non-meromorphic first integrals, integrability on a domain disjoint from the obstructed neighborhood, or a different notion of integrability. It does not cover zero angular momentum, nonzero Ξ magnetic dynamics, gauge-dependent or clock-dependent functions, or triple-collision regularization.

The computation begins with a published reduced Hamiltonian convention. It checks exact equivalence to that convention but does not independently reconstruct every canonical translation and rotation map from the unreduced eighteen-dimensional state. The spatial transfer is an explicit differential subquotient argument; it is not the invalid inference that nonintegrability of an invariant subsystem alone implies nonintegrability of the ambient system.

No global orbit classification follows. The only domain-level corollary is by restriction: a complete admissible family on a larger domain containing the obstructed regular neighborhood would restrict to a forbidden local family.

9 Reproducibility

The repository contains the manuscript source, bibliography, claim ledger, proof-dependency graph, canonical artifact manifest, exact predecessor hashes, and the source-disjoint SymPy reconstruction. From a clean checkout, the core package is exercised by

```
./scripts/reproduce_wp5_pub_v1.sh --core
```

The command verifies the specification lock and all pinned inputs, rebuilds the PDF, reruns the generic symbolic derivation, compares its complete decision-bearing payload after removing elapsed time, executes the publication-specific adversarial tests, and emits a machine-readable report. A full mode additionally runs the partitioned legacy baseline.

Primary papers are identified by bibliographic metadata, URL, and SHA-256 download hash but are not redistributed. External review is a separate, fail-closed gate: two independent human reviewers must collectively cover the planar/differential-Galois and symplectic-reduction/KS arguments, and at least one clean-room reproduction must occur outside project-controlled hardware. No internal process, automated agent, or self-review is treated as that evidence.

10 Conclusion

The package reproduces the published planar Morales–Ramis calculation from the reduced Hamiltonian and verifies its positive-mass non-Abelian Galois obstruction with exact arithmetic. A separately qualified subquotient, cotangent-reduction, Sundman, and shell-rank argument transfers that obstruction to the translation-reduced spatial Hamiltonian and to the declared KS atlas. This establishes the local, complete-meromorphic nonintegrability statement in Theorem 6, subject to every stated model boundary. External mathematical review, independent clean-room reproduction, author metadata, license choices, and release authorization remain publication gates rather than inferred facts.

References

[1] Delphine Boucher and Jacques-Arthur Weil. Application of the Morales–Ramis theorem to test the non-complete integrability of the planar three-body problem. Author-hosted primary manuscript, source PDF.

[2] Paul E. Kustaanheimo and Eduard L. Stiefel. Perturbation theory of Kepler motion based on spinor regularization. *Journal für die reine und angewandte Mathematik*, 218:204–219, 1965. DOI record.

[3] Xiaoming Li, Xin Li, and Shijun Liao. One family of 13315 stable periodic orbits of non-hierarchical unequal-mass triple systems. *Science China Physics, Mechanics & Astronomy*, 2021. DOI record.

[4] Jerrold Marsden and Alan Weinstein. Reduction of symplectic manifolds with symmetry. *Reports on Mathematical Physics*, 5(1):121–130, 1974. DOI record.

[5] Juan J. Morales-Ruiz and Jean-Pierre Ramis. Galoisian obstructions to integrability of Hamiltonian systems i. *Methods and Applications of Analysis*, 8(1):33–96, 2001. DOI record.

[6] Alexei V. Tsygvintsev. The meromorphic non-integrability of the three-body problem. *Journal de Mathématiques Pures et Appliquées*, 81, 2002. arXiv:math/0009218.